Lab 6 · Case Study: A Cluster on the Surgical Ward

Day 2 · 13:15–15:00 · 105 min · Groups of 4 Objective: LO3–LO5 — Combine genomic and epidemiological evidence to decide whether an outbreak is happening, and communicate the decision

> ⚠️ **About this case:** the genomes are **real**: MRSA isolates from a published hospital outbreak investigation. The hospitals, wards, patients and dates are **fictional**, written so the case feels like our setting. The real story is revealed at the end.

The situation

District Hospital A (fictional) · Monday, 16 March 2026

The infection prevention and control (IPC) nurse writes to you:

> "Since early February we've had **more MRSA wound infections than usual on Surgical Ward B**. The surgeons say it's bad luck. Theatre staff say it's the ward. The ward says it's the theatre. The hospital director asks whether we should **close Ward B to admissions**. Closing it would cancel about 40 operations a week. The lab stored all MRSA isolates since September, and we got genomes back from our sequencing partner last week. **Can you tell us what's going on by Wednesday's IPC meeting?**"

Your job: answer one question for the director: **should we close Ward B?**

Your evidence

1. **Your Pathogenwatch collection** from Lab 5: tree, timeline and metadata (ward, theatre, dates) in one place
2. **SNP distance matrix**: `koser_snp_matrix.tsv` (in `labs/data/`). Open it in Excel.

Group roles

Role	Job
Genomics lead	Reads the tree and SNP matrix
Epidemiologist	Reads the timeline and metadata: who was where, when
Clinician / IPC	Thinks through how MRSA could spread, and what to do
Communicator	Writes and delivers the 5-minute briefing

Part A — Look at the cases (15 min)

Open your Pathogenwatch collection. Use the **timeline** and the **data table**.

✍️ Questions:

1. How many MRSA isolates in total? How many from **Surgical Ward B since 1 February**?
2. On the routine **lab report**, every isolate is MRSA and cefoxitin-resistant. Does that mean they are all one outbreak?

Part B — Find the cluster (25 min)

B1. The tree

In Pathogenwatch, look for a group of isolates on very **short branches**. Write down their patient codes.

B2. The SNP matrix

Each cell is the number of genetic differences (SNPs) between two isolates. **Small number = closely related.**

1. In Excel, select the numbers → **Conditional Formatting** → **Highlight Cells Rules** → **Less Than** → `100`
2. Look at the highlighted cells. You should see a clear **gap**: some isolates are all close to each other, and everything else is far away.

Cluster isolates (patient codes)	Largest distance inside the cluster	Smallest distance to an isolate outside it

> 💡 **Why look for a gap?** In L4 you saw rough thresholds for *S. aureus* (about 15–30 SNPs). They are guides, not laws. In real data, the clearest sign is a group that is close together with a big gap to everything else.

✍️ Does the cluster from the SNP matrix match the short-branch group on the tree?

Part C — Genomics + epidemiology together (20 min)

Colour the tree by **ward**, then by **theatre**. Then fill in the table for each **Ward B patient since February**. The first row is done for you.

Patient	In the cluster?	Theatre	Linked to the outbreak?
P01	Yes	Theatre 2	**Linked**
P02			
P03			
P05			
P06			
P07			
P08			

✍️ **Three questions for your group:**

1. **Are all the Ward B cases one outbreak?** Look closely at **P08**: same ward, same time, same lab result. Is P08 in the cluster?
2. **Is anyone in the cluster *not* on Ward B?** What do they have in common with the Ward B cases?
3. **So, should we close Ward B?** Where is the outbreak most likely coming from?

🚀 **If you finish early:**

- Do any isolates from **other hospitals** belong to the cluster?
- **P07** is in the cluster but a bit further from the others. Why might one isolate collect extra mutations faster?
- The cluster carries a mupirocin-resistance gene (see your Lab 5 AMR view). Why does that matter if a **staff carrier** is found and needs decolonising?

Part D — Prepare the briefing (25 min)

Prepare a **5-minute briefing** for the IPC committee (director, chief surgeon, nursing lead, lab head). They are **not** genomics experts.

Use this template (one flip-chart page or 3 slides):

1. Bottom line (1 sentence): "We found / did not find evidence of an outbreak of _____ linked to _____." **2. What we did:** "We sequenced __ MRSA isolates from __ to __ and compared them genetically." **3. What we found:** one visual (screenshot of the Pathogenwatch tree + timeline) and 2–3 bullet points **4. What this means:** which cases are linked, and the likely route **5. Recommendations:** 3 concrete actions, with who does them and by when **6. What we don't know yet:** limitations and next steps

Rules: no jargon without explanation ("SNP" → "genetic differences"), and no more than **one** figure.

Part E — Briefings (15 min, plenary)

Each group gets **5 minutes** to present to the "IPC committee" (the instructor + one participant group acting as committee members). The committee must ask at least one question:

- "*Should we close Ward B?*"
- "*Is it the surgeons' fault?*"
- "*How sure are you?*"
- "*What will this cost?*"

The reveal (instructor, 5 min)

The genomes come from **Köser et al., *N Engl J Med* 2012**: an investigation of a suspected MRSA outbreak in a **special care baby unit** (neonatal unit) in Cambridge, UK. Rapid whole-genome sequencing distinguished the outbreak isolates from unrelated MRSA that looked identical on routine lab tests, and showed transmission that conventional methods had not resolved. A follow-up study (Harris et al., *Lancet Infect Dis* 2013) used genomics to trace transmission into the community and identify a **staff carrier**, which ended the outbreak.

Take-home messages:

1. **Same ward + same lab result ≠ same outbreak.** Genomics separates linked from unlinked cases.
2. **Genomics finds hidden links** (shared theatre, staff, equipment) that ward-based thinking misses.
3. **Genomics can prevent unnecessary actions** (closing a ward) as well as trigger necessary ones.
4. Genomic results are only useful **with good metadata** and **clear communication**.

Genomic data: ENA project PRJEB2912 (Köser et al. 2012). Scenario metadata are fictional and created for teaching.